

Syllabus

MAT 766 Variable Selection, Model Validation, and Nonlinear Regression

Instructor

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Course Description

If you have a technical background in mathematics/statistics/computer science/engineering and or are pursuing a career change to jobs or industries that are data-driven, this course is for you. Those industries might be finance, retail, tech, healthcare, government, or many others. The opportunity is endless.

This course will focus on getting you acquainted with the generalized linear model (GLM) through the examples of logistic and Poisson regression. You will also see how simple and multiple linear regression relates to GLM using the link function. We will also study a regression technique that is robust to having outliers in the data. Finally, we will learn how to perform model validation involving GLM.

Course Outcomes

After taking this class, students will be able to:

- Determine which regression models to use based on the nature of the response variable.
- Use regression technique which is robust to the presence of outliers.
- Perform generalized linear regression using R by identifying the correct link function.
- Interpret and draw conclusions on the regression model.
- Use R to perform statistical inference based on the regression models.

Course Materials

The link to reading materials and resources to learn on the topics can be found in each week's learning module. All materials are available online for free, no required resources need to be purchased. There is no required textbook to supplement the course materials. Note: Be aware that some resources may open in a new tab.

Software Requirements: R Studio

This course requires the use of RStudio. The RStudio integrated development environment is a set of tools built to help you be more productive with R and Python.

You may download RStudio here: <https://posit.co/download/rstudio-desktop/>.

Course Outline

The course consists of 3 modules that focus on the following key areas:

Module 1: Logistic Regression

Key concepts

- Lesson 1: Introduction to Logistic Regression, Part I
- Lesson 2: Introduction to Logistic Regression, Part II

Readings

- Slides will be available for download and review pertaining to each of the corresponding lessons contained within the module.

Module 2 Poisson Regression and Generalized Linear Model

Key concepts

- Lesson 1: Poisson Regression
- Lesson 2: Generalized Linear Model

Readings

- Slides will be available for download and review pertaining to each of the corresponding lessons contained within the module.

Module 3 Robust Regression and Model Validation

Key concepts

- Lesson 1: Robust Regression
- Lesson 2: Model Validations

Readings

- Slides will be available for download and review pertaining to each of the corresponding lessons contained within the module.

Course Structure and Learning Activities

There are 3 content modules in this course and each module may take about 6-7 hours to complete. Each module represents 1 week of learning with corresponding assignments per module. You can advance at your own pace; it is recommended that you take no longer than a week to complete each module. A consistent pace will help you complete the module and move on to the next course in the sequence. The final module consists of your final exam for the course.

This course is comprised of the following elements:

- **Readings:** Each module may include several required and/or supplemental readings.
- **Video Lessons:** In each module, the concepts you need to know will be presented through a collection of short videos. You may stream these videos for playback within the browser by clicking on their titles.
- **Discussion Forum:** This course has a place for you to interact with other learners about class-related

topics. Unless specified, discussion forums do not carry a score.

- **Practice Quizzes:** Each module will include some practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.
- **Summative Module Assessments:** Each module will include at least one summative module assessment. You will be allowed one attempt every eight hours for each assessment. There is no time limit on how long you take to complete each attempt at the assessment. Your highest grade will be recorded
- **Final Assessment:** This course will contain one final summative course assessment. Before taking the exam, please make sure you are in a place with reliable internet connection. No retakes will be granted for the lack of internet access. You are in an online program and the use of the Internet is a requirement.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. Only those who complete and submit all assignments, including peer reviews, will receive a certificate of completion of this course. *No late assignments will be accepted.* In case of extenuating circumstances beyond your control that prevent the submission of an assignment or exam, you have to enter a request with the program advisor and the instructor.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera [Financial Aid](#) is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, [please see the help topics here](#).

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	3-6 per module	.5-1 hour	N/A	N/A
Practice Quizzes	No	3-6 per module	.5 hour	N/A	N/A

Discussions	No	2 per course	1 hour	N/A	N/A
Summative Module Assessments	Yes	1 per module	.5 hour	80%	20%/each module (60%)
Summative Course Assessment	Yes	1 per course	1-3 hours	80%	40%

Getting and Giving Help

- Use the [Learner Help Center](#) to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with [Coursera's policy on Accessibility](#).

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to [Coursera's Honor Code](#), we also have high expectations for conduct during course participation.

Discussion Forums: Expectations

Sharing an online course with other avid learners like you gives you a unique opportunity to share, collaborate, and learn from others and their experiences, and helps you reinforce your understanding of the topics of the course. Interacting in the Discussion Forums is a great way to engage with your online community. We know that it is not possible to read every discussion forum post, so we recommend that you read those that interest you; and reply when you can contribute. The forum is part of your class activities and everybody is expected to act professionally and be civil and respectful of others in your class. Failure to meet these expectations may be considered a break in the Academic Code of Conduct and may result in your removal from the course. Please, check tips and helpful tools to [interact in discussion forums in this document](#).

Academic Code of Conduct

Above all else, learners are expected to ensure that their conduct helps to create an atmosphere conducive to learning and the interchange of knowledge. While it is understood that some of these items are subject to interpretation, learners should nonetheless endeavor to:

- Be respectful of fellow learners.
- Not discriminate against fellow learners in any manner.

- Conduct peer reviews in a timely manner and give useful feedback on what was done well, helpful suggestions for how to improve, and specific comments about why you gave the grade you chose to assist peers in their learning.
- Turn assignments in on time and follow instructions on all assignments including those affecting the use of technology.
- Be truthful in all communication, which includes, but is not limited to, avoiding academic dishonesty.

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